Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/31/2015
PI Name:	Kamotani, Yasuhiro Ph.D.		
Project Title:	Cyclonic Two-Phase Flow Separator Experiment in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	yxk@case.edu	**Fax:**	FY 216 368-6445
PI Organization Type:	UNIVERSITY	**Phone:**	216 368-6455
Organization Name:	Case Western Reserve University		
PI Address 1:	Department of Mechanical and Aerospace Engineering		
PI Address 2:	10900 Euclid Ave, Glennan 412		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2014	**End Date:**	05/30/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Sharp, Lauren	**Contact Phone:**	
Contact Email:	lauren.m.sharp@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan Ph.D. (Case Western Reserve University)		
Grant/Contract No.:	NNX14AJ74G		
Performance Goal No.:			
Performance Goal Text:			

Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components.
Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems.

Task Description:

It is a lack of basic engineering and scientific knowledge that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators. Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using computational fluid dynamic (CFD) is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues.

The main objectives are to:

1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct.

2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators.

3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity.

4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity.

To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities.

The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to:
1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems

2. Provide details of the columnar stability of the gas core

3. Provide details of the time constants associated with the response of passive cyclonic separators

4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity

The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity.

Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.

The objective of the TPFSE (Two-Phase Flow Separator Experiment) project is to facilitate the maturation of passive cyclonic separation technology through exhaustive research of a newly designed gas-liquid phase separator evolved from NASA Glenn Research Center's (GRC) existing Cascade Cyclonic Separation Device (CSD-C). Combined experimental and numerical analyses are being performed to quantitatively assess and delimit the range of operability of the new separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct – features that have yet to be completely explored in any such device.
The approach is first to assess the range of operability of the new separator design, the Case Western Reserve University Separator, through the careful construction of detailed flow separation maps using data gathered from reduced-gravity flight experimentation in conjunction with ancillary computational studies. Second, the stability behavior of the device will be observed and analyzed. Special attention will be paid both to device-level conduct and also to behaviors that could manifest dysfunction elsewhere at the complete system level. A hybrid multifluid-VOF computational multiphase fluid dynamics (CFD) method has been conducted in order to permit discernment of these flow features. Lastly, with the ensuing understanding of the causes of any performance-affecting responses, a phenomenological description of the separator will be sought; in particular, the insight acquired should facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new designs.

Large-Eddy Simulation of Turbulent Multiphase Flows within Cyclone Separator

A hybrid multiphase method based on the combination of an Eulerian multifluid framework and VOF sharp interface capturing has been continuously improved using libraries and utilities of OpenFOAM to construct a custom CFD solver. Turbulence is treated using Large Eddy Simulation (LES) with the Smagorinsky sub-grid model. Our earlier

Task Progress:	experimental work shows that flow features within separators are three dimensional and highly time dependent multiphase flow. LES has the advantage of computing the unsteady three dimensional flow features which are necessary to accurately predict flow quantities. To save computational time, a 2D-axisymmetric CFD simulations has been conducted to validate our custom solver. It has been demonstrated from our preliminary results that the LES model is capable of capturing instantaneous eddy formation and flow features, and hence can provide more accurate and detailed flow motion within separators. Typical results for single and multiphase injection into a standard separator design with 2D-axisymmtric model. Moreover, a technique of local switching sharpening interface is being developed to improve the way to couple between eulerian-eulerian and VOF methods. The interface sharpening which is turned on or off is dependent on local gradient of the volume fraction. Where the interface exists (steep gradient in volume fraction), then interface sharpening is turned on to apply interface compression, and where it is dispersed, is set to off to let it stay dispersed. Eventually, a three-dimensional and time-dependent large eddy simulation for our separator will be performed. Gas-Rotating Liquid Interface Instability It is important to have a stable gas core for the effective operation of cyclonic separators. However, there are some conditions under which the gas core collapses. A CFD and control-volume-based analytical approach has been performed to study the dynamic behavior of the gas core in microgravity environment. In the absence of buoyancy force, the forces that affect the interface are centrifugal and surface tension (capillary) forces. The centrifugal force makes the interface stable, while the surface tension force destabilizes the interface and breaks down the gas core into gas bubbles. Our earlier control volume analysis, based on the pressure balance and the conservation of angular momentum, predicts the average liquid layer thickness (or the gas core radius) and the average angular velocity in the separator. The analysis shows that the important parameters for the flow are: the injection Reynolds number (Re), the separator aspect ratio (As =length/radius), the liquid outlet back pressure coefficient (K), the area ratio (G=separator cross-section/inlet nozzle cross-section), the swirl number (S), and the inlet gas volumetric quality (ß). This control-volume-based analytical approach (initially developed in previous years) has been modified to include the surface tension for the present work, which introduces the Weber number (We). The resulting ordinary differential equations are solved numerically, for given parameters, to determine the film thickness and the gas core size as before. When We becomes smaller than a certain value, the problem has no solution, which gives the critical We (Wecr), for given Re, As, K, and G, for the collapse of the gas core. The results are shown the values of Wecr are not sensitive to any of those parameters, namely the gas core collapses around We ˜ 0.7 for any condition. Our investigation reveals that surface tension effects can offset the basic pressure balance for the separator and break-up the core. When the pressure drop due to surface tension exceeds a certain value, the pressure balance within separators cannot be maintained and the liquid film collapses. Supporting Work for Flight Hardware Design A breadboard model of the proposed experiment is being designed and built at NASA/GRC. We are in constant contact with NASA/GRC engineers for the design and work with NASA/GRC engineers to finalize the Science Requirements Document for the experiment. For these experiments, it is required to measure gas core diameter as well as void fraction distribution in liquid annulus reside at side of the separator. An electrical capacitance volume tomography (ECVT) system is proposed to use. ECVT is a recently developed technology by Tech4Imaging providing 3-D volume image measurements in a separator. Some preliminary data show that ECVT might be capable for our requirements. To convert the capacitance to void fraction value, a calibration is required. To do so, classical method quick-closing valves was used to calibrate a concave plate capacitance sensor. The calibration flow loop had been constructed to conducting the measurement of void fraction and calibration at Case Western Reserve University (CWRU) side. The preliminary calibration results of development pipe had been taken. The results show that the sensors had a linear response to a void fraction range of 0 to 0.7. However, an offset phenomenon for capacitance measurements with two phase flow and some issues for extracting accurate gas core size and void fraction distribution by utilizing all available capacitance measurements had been observed. We are collaborating with NASA/GRC and Tech4Imaging teams to resolve this issue.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Kang MF, Kharraz AO, Kadambi J, Kamotani Y. "Investigation of Passive Cyclonic Two-Phase Separator Performance in Microgravity." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22- 26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014